

Module 05: Action in Robotics

Learning Objectives

1. Define **Action** within the context of Robotics.
2. Analyze how Action interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Action.

Introduction

This module explores **Action**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Action is a critical component of the 8-part Active Inference spine, bridging the gap between Cognition and Learning.

Key Concepts

1. Action as a Markov Blanket Boundary

How does Action define the boundary between the agent and the environment?

2. Generative Models of Action

What parameters involved in Action must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Action drive the perception-action loop?

Applications

In Robotics, we see Action manifest in: * **Specific Example 1:** An autonomous race car (such as those in the Indy Autonomous Challenge) executes high-speed overtaking maneuvers at

250+ km/h by generating action commands through a nonlinear MPC that minimizes expected free energy over a 2-second horizon: the pragmatic component drives the car toward the optimal racing line while the risk-aversion component penalizes trajectories that come within unsafe distances of competitors, with tire-force saturation limits encoded as hard constraints on the action space that the generative model must respect. * **Specific Example 2:** An autonomous agricultural robot performing selective weeding uses a delta-mechanism actuator to precisely strike individual weeds detected by an overhead camera at 20 targets per second; each strike action is generated by minimizing the prediction error between the desired impact point (weed centroid from vision) and the predicted tool tip position (from the delta kinematic model), with the active inference loop running at 200 Hz to compensate for plant sway due to wind and vehicle vibration during field operation.

Conclusion

Understanding Action allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.